First Order Differential Equations

Prof. A. K. B. Chand

Indian Institute of Technology
Madras

28-7-2026



Introduction

- An ordinary differential equation (ODE) is an equation involving an unknown function of a single independent variable and one or more of its derivatives.
- The **order** of an ODE is determined by the highest-order derivative appearing in the equation.
- The **degree** of an ordinary differential equation is the power (exponent) of the **highest-order derivative** appearing in the equation, after the equation has been made free of:
  - fractional powers or roots of derivatives
  - negative powers of derivatives
  - derivatives appearing in denominators (fractions)

  In other words, first write the ODE as a polynomial equation in derivatives; the degree is then the exponent on the highest derivative term.



Examples

Example 1

$$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + y = 0$$

Order $= 2$ (highest derivative is the 2nd derivative)
Degree $= 3$ (that highest derivative is raised to the power 3)

Example 2

$$\frac{d^2y}{dx^2} + \sqrt{\frac{dy}{dx}} = 0$$

Here $\frac{dy}{dx}$ appears under a square root, thus not in polynomial form.
Squaring to remove the radical:

$$\left(\frac{d^2y}{dx^2}\right)^2 = \frac{dy}{dx}$$

Order $= 2$, Degree $= 2$



Examples

Important notes

- **Order** = highest derivative present (always well-defined).
- **Degree** is only defined when the ODE can be expressed as a polynomial in its derivatives. If not (e.g., derivatives inside sin, log, or exponential functions), the degree is **not defined**.

Example where degree is undefined

$$\frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$$

Order $= 2$, but degree is undefined since $\frac{dy}{dx}$ sits inside a sine function.



First Order ODEs

- A first-order ordinary differential equation is an equation in which the highest derivative of the unknown function is the first derivative. It can be expressed in the general form

$$F(x, y, y') = 0,$$

where x is the independent variable, $y = y(x)$ is the dependent variable, and $y' = \frac{dy}{dx}$.

- If the equation can be solved explicitly for the derivative, it takes the form

$$\frac{dy}{dx} = f(x, y),$$

which is known as the explicit form of a first-order ODE.

- When the derivative cannot be isolated explicitly, the equation is said to be in implicit form.
- The objective in solving a first-order ODE is to determine a function $y(x)$ that satisfies the differential equation on a given interval.



- Explicit form : $\frac{dy}{dx} = 3x + 2y$.
- Implicit form : $x\left(\frac{dy}{dx}\right)^2 + 7y = 4$.
- Differential form: $M(x,y)dx + N(x,y)dy = 0$, where M and N are continuous functions.

Definition

A **general solution** of a first-order differential equation is a family of solutions containing one arbitrary constant.

Example: Consider

$$\frac{dy}{dx} = 2x.$$

Integrating both sides,

$$\boxed{y = x^2 + C}, \quad C \text{ is an arbitrary constant}$$

is the general solution. Each value of C produces a different solution curve.



Particular Solution and IVP

- A particular solution is obtained by assigning a specific value to the arbitrary constant.
- For $\frac{dy}{dx} = 2x, \qquad y(1) = 5$, the particular solution is obtained from the general solution $y = x^2 + C$ using the given condition.
- Here $5 = 1 + C \implies C = 4$ gives the particular solution as $\boxed{y = x^2 + 4}$.

Definition

A first-order differential equation together with an initial condition

$$y(x_0) = y_0$$

is called an **initial value problem (IVP)**.

$$\boxed{\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0.}$$



Geometric Interpretation

- The general solution represents a family of curves.
- The initial condition selects the unique curve passing through the specified point.
- Thus, solving an IVP means finding the integral curve satisfying $y(x_0) = y_0$.

Many scientific phenomena are naturally described by first-order differential equations.

Population Growth: Suppose $P(t)$ denotes the population at time t. If the rate of growth is proportional to the population,

$$\frac{dP}{dt} = kP,$$

where $k > 0$ is the growth constant. The solution is

$$P(t) = P_0 e^{kt}.$$



Applications

Radioactive Decay: If the decay rate is proportional to the remaining quantity,

$$\frac{dN}{dt} = -\lambda N, \quad \lambda > 0.$$

The solution is

$$N(t) = N_0 e^{-\lambda t}.$$

Newton's Law of Cooling: Let $T(t)$ denote the temperature of an object. Newton's law states

$$\frac{dT}{dt} = -k(T - T_s),$$

where T_s denotes the surrounding temperature. The solution is

$$T(t) = T_s + (T_0 - T_s)e^{-kt}.$$



Isoclines

- For a first-order ODE

$$\frac{dy}{dx} = f(x, y),$$

an **isocline** is a curve along which every solution curve has the same constant slope c.

- It is defined implicitly by

$$f(x, y) = c.$$

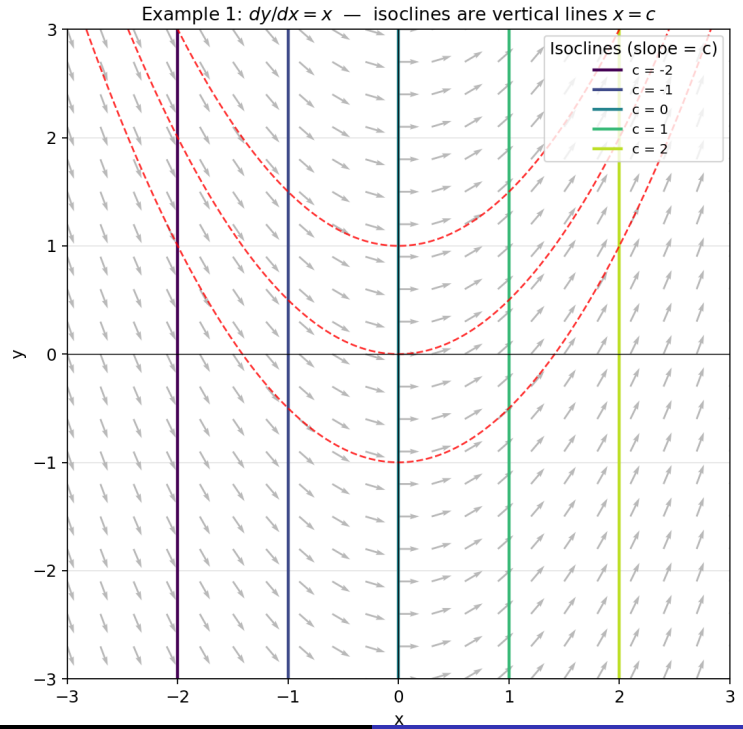
- Isoclines do not solve the ODE, but they are a graphical tool.
- By drawing several isoclines (for different values of c) and sketching short line segments of slope c along each one, we build up the *direction field*, which allows us to sketch approximate solution curves without integrating the equation.
- This technique is usually introduced before Euler's method or separation of variables, since it builds geometric intuition for the shape of solutions.



Example-1

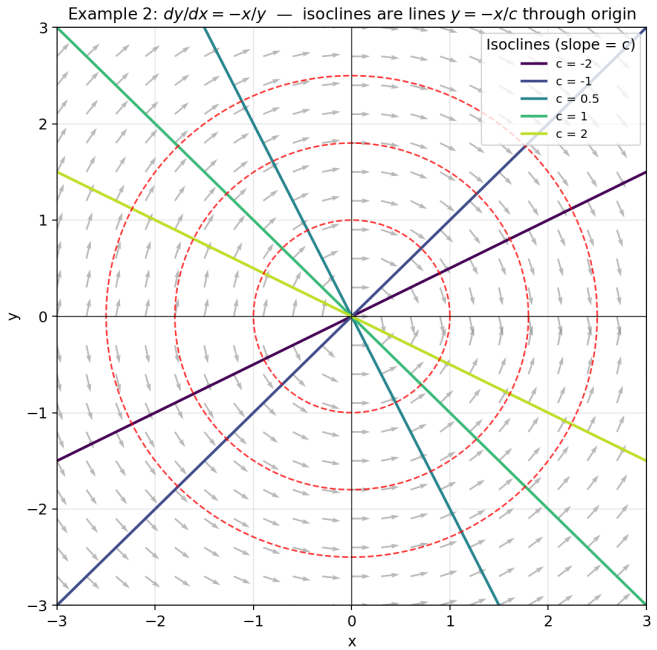
- For $\dfrac{dy}{dx} = x$, $f(x, y) = x$ depends only on x, so the isoclines are the **vertical lines** $x = c$.
- Along the line $x = c$, every solution has slope exactly c.
- The general solution is

$$y = \frac{x^2}{2} + K (\text{a family of parabolas}).$$



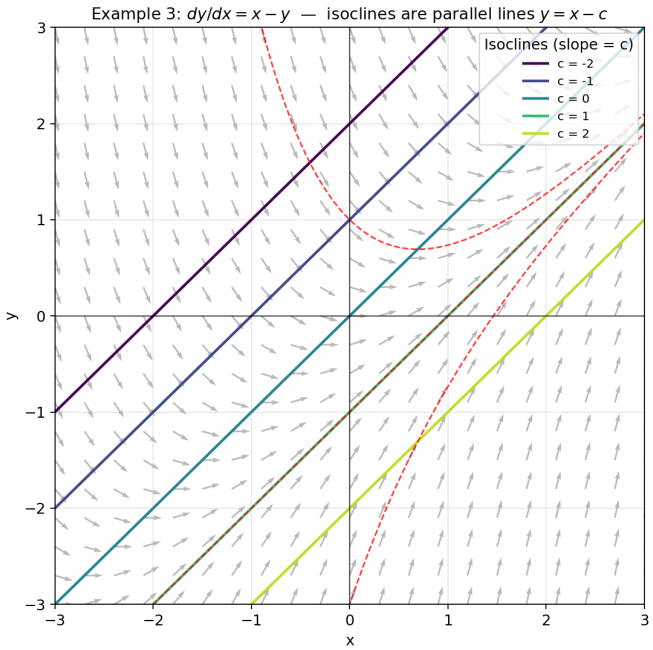
Example-2

- For $\frac{dy}{dx} = -\frac{x}{y}$, setting $f(x, y) = -x/y = c$ gives $y = -\frac{x}{c}$, so the isoclines are **straight lines through the origin**.
- Solving the ODE by separation of variables ($y\,dy = -x\,dx$) gives a family of concentric circles : $x^2 + y^2 = r^2$.
- Notice that the isoclines (radial lines) are *not* the solution curves (circles) - a clear illustration that isoclines only mark constant slope, not the trajectories themselves.



Example-3

- For $\dfrac{dy}{dx} = x - y$, taking $f(x, y) = x - y = c$ gives $y = x - c$, so the isoclines are **parallel lines of slope 1**.
- This is a linear ODE, $y' + y = x$, with general solution $y = x - 1 + Ke^{-x}$.
- As $x \to \infty$, all solution curves approach the line $y = x - 1$.



Existence and Uniqueness (Picard–Lindelöf Theorem)

Consider the initial value problem (IVP):

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0$$

Two natural questions arise:

1. **Existence** — Does a solution exist at all?
2. **Uniqueness** — If a solution exists, is it the *only* one?

1. Existence: Let $f(x, y)$ be defined and *continuous* on the rectangle

$$R = \{(x, y) : |x - x_0| \leq a,\ |y - y_0| \leq b\}.$$

Then, the IVP has solution(s).



2. Uniqueness: Suppose further that f satisfies a **Lipschitz condition** in y on R, i.e. there exists a constant $L > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq L\,|y_1 - y_2| \qquad \text{for all } (x, y_1), (x, y_2) \in R.$$

Then there exists a **unique** solution $y(x)$ of the IVP on the interval $|x - x_0| \leq h$, where

$$h = \min\left(a, \frac{b}{M}\right), \qquad M = \max_{(x,y)\in R} |f(x, y)|.$$

Note that if $\dfrac{\partial f}{\partial y}$ exists and is continuous (hence bounded) on R, then f automatically satisfies a Lipschitz condition on R (**Stronger condition than Lipschitz Condition**).



Existence and Uniqueness

- Consider the IVP $y' = y, \qquad y(0) = 1.$
- Here $f(x, y) = y$ is continuous everywhere, and $\dfrac{\partial f}{\partial y} = 1$ is continuous, so the Lipschitz condition holds.
- The P-L theorem guarantees a unique solution: $y(x) = e^x$.
- Consider another IVP: $y' = y^{1/3}, \qquad y(0) = 0.$
- Here $f(x, y) = y^{1/3}$ is continuous, so a solution exists. However,

$$\frac{\partial f}{\partial y} = \frac{1}{3} y^{-2/3} \to \infty \quad \text{as } y \to 0,$$

  so the Lipschitz condition **fails** near $y = 0$.
- Consequently this IVP has **infinitely many** solutions, including $y(x) = 0 \qquad$ and $\qquad y(x) = \left(\frac{2x}{3}\right)^{3/2}$.
- This demonstrates that continuity alone is insufficient for uniqueness; the Lipschitz condition is essential.



Exact Differential Equations

Definition

The differential form $M(x, y)\,dx + N(x, y)\,dy$ is said to be **exact** in a region R of the xy-plane if there exists a function $F(x, y)$ such that

$$dF = \frac{\partial F}{\partial x}\,dx + \frac{\partial F}{\partial y}\,dy = M\,dx + N\,dy$$

throughout R; that is, $F_x = M$ and $F_y = N$. The equation $M\,dx + N\,dy = 0$ is then called an **exact equation**.

- Since $M\,dx + N\,dy$ is the total differential of some function F, then the equation $M\,dx + N\,dy = 0$ reduces to $dF = 0$, whose solution is immediately $F(x, y) = C$.



Test for Exactness

Theorem

Let $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives in a simply connected region R. Then $M\,dx + N\,dy = 0$ is exact in R if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \qquad \textit{throughout } R.$$

Sketch of proof

($\Rightarrow$) If the equation is exact, then $M = F_x$ and $N = F_y$ for some F. Since F has continuous second partials, the mixed partials are equal:

$$\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial y\,\partial x} = \frac{\partial^2 F}{\partial x\,\partial y} = \frac{\partial N}{\partial x}.$$



Sketch...

($\Leftarrow$) Conversely, if $M_y = N_x$, define

$$F(x, y) = \int M(x, y)\, dx + g(y),$$

where the integration is performed treating y as constant, and $g(y)$ is an unknown function of y alone.
Differentiating with respect to y and setting the result equal to N determines $g'(y)$.
Here the condition $M_y = N_x$ guarantees that the resulting expression for $g'(y)$ depends only on y, so $g(y)$ can be found by ordinary integration.
This constructs the required F.
Note that one can take $F(x, y) = \int N(x, y)\, dy + h(x)$ and proceed to compute $h(x)$.



Exact DE- Example 1

- Solve $(2xy + 3)\,dx + (x^2 - 1)\,dy = 0$.
- Here $M = 2xy + 3$, $N = x^2 - 1$. To verify the exactness:

$$M_y = 2x, \qquad N_x = 2x \quad \Rightarrow \quad M_y = N_x.$$

- The equation is exact. Integrate M with respect to x:

$$F = \int (2xy + 3)\,dx = x^2y + 3x + g(y).$$

- Differentiate with respect to y and equate to N:

$$F_y = x^2 + g'(y) = x^2 - 1 \quad \Rightarrow \quad g'(y) = -1 \quad \Rightarrow \quad g(y) = -y.$$

- Hence the general solution is

$$x^2y + 3x - y = C.$$



Exact DE - Example 2

- Solve $(y\cos x + 2xe^y)\,dx + (\sin x + x^2e^y - 1)\,dy = 0$.
- Here $M = y\cos x + 2xe^y$, $N = \sin x + x^2e^y - 1$. Check:

$$M_y = \cos x + 2xe^y, \qquad N_x = \cos x + 2xe^y \quad \Rightarrow \quad M_y = N_x.$$

- The DE is Exact. Integrating M with respect to x:

$$F = \int (y\cos x + 2xe^y)\,dx = y\sin x + x^2e^y + g(y).$$

- Differentiating with respect to y:

$$F_y = \sin x + x^2e^y + g'(y) = \sin x + x^2e^y - 1$$

$$\Rightarrow \quad g'(y) = -1 \Rightarrow g(y) = -y.$$

- General solution:

$$y\sin x + x^2e^y - y = C.$$



Exact DE - Example 3

- Solve $e^y\,dx + (xe^y + 2y)\,dy = 0$.
- Here $M = e^y$, $N = xe^y + 2y$ and $M_y = e^y = N_x$ confirm the exactness.
- Thus, there exists a function $F(x, y)$ such that $F_x = e^y$ and $F_y = xe^y + 2y$.
- Integrating first equation with respect to x gives $F = \int e^y\,dx = xe^y + g(y)$.
- Differentiating the above equation with respect to y and equating with 2nd equation, $g'(y) = 2y \implies g(y) = y^2$.
- The general solution is $F(x, y) \equiv xe^y + y^2 = C$.
- Alter: If we start with 2nd equation and integrate w.r.t y, we have $F = xe^y + y^2 + h(x)$.
- Differentiating the above equation with respect to x and equating with 1st equation, $h'(x) = 0 \implies h(x) = C_1$.
- The general solution is $F(x, y) \equiv xe^y + y^2 + C_1 = C$ which is equivalent to $xe^y + y^2 = K$, where $K = C - C_1$.
  (Caution: Don't write two constants in the final solution)

